

UD65640

Application Note

Final Test for FPC Trim OSC

AN-K0018-250715

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1. Target

Test UD65640 CMOS final test for trim OSC, check IC Status is normal.

Test List

No	Item	Description	Use Pad Name
1	OSC Frequency Trim	Trim GPIO1 pad single at 3.125MHz ~ 3.135Mhz.	GPIO1
2	OTP Program	Programmer trim value and check whether the reload OTP is normal	

Table 1

2. Register

Access Control: Manufacturer Command Access Control (80h)

80h	Access Control									
	W/R	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	—	1	0	0	0	0	0	0	0	80h
Parameter 1	W/R	0	0	0	0	0	0	0	MCS_EN	00h
Descriptions	MCS_EN: This command releases parameters so that Manufacturer Command Access are enabled. 1: Access enabled. 0: Access denied. (Protection on) 1 = Must fill 1; 0 = Must fill 0; X = Don't care.									
Restrictions										

Device Code: Device Code (88h)

88h	Device Code									
	W/R	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	—	1	0	0	0	1	0	0	0	88h
Parameter 1	W/R	DDB0[7:0]								00h
Parameter 2	W/R	DDB1[7:0]								00h
Parameter 3	W/R	DDB2[7:0]								00h
Descriptions	DDB0: This command is for user command (DAh) version number. DDB1: This command is for user command (DBh) version number. DDB2: This command is for user command (DCh) version number. 1 = Must fill 1; 0 = Must fill 0; X = Don't care.									
Restrictions										

Analog Settings: Analog Settings (90h)

90h	Analog Setting									
	W/R	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	—	1	0	0	1	0	0	0	0	90h
Parameter 1	W/R	DV_VREFP_OPT [3:0]				0	DV_TRIM_DVDD [2:0]			70h
Parameter 2	W/R	0	DV_TRIM_OSC [6:0]							40h
Parameter 3	W/R	0		DV_BG_TRIM [5:0]						00h
Parameter 4	W/R	0				DV_BG_TRIM_TC [3:0]				00h
Descriptions	DV_VREFP_OPT [3:0]: VREFP Voltage Output Set.									
	DV_TRIM_DVDD [2:0]: Trim DVDD Voltage.									
	DV_TRIM_OSC [6:0]: Trim OSC Frequency.									
	DV_BG_TRIM [5:0]: Trim BG Voltage.									
	1 = Must fill 1; 0 = Must fill 0; X = Don't care.									
Restrictions										

3. Final Test Flow:

3.1 OSC Frequency Trim

Step1. Register write 11h

Step2. Register write 29h

Step3. Register write 80h, 01h

Step4. Register write F8h, 00h, 22h

Step5. Register write 35h, 00h

Step6. Measure GPIO1 pin, frame rate target 3.125Mhz ~ 3.135MHz.

Step7. Register adjustment 90h.P2 DV_TRIM_OSC [6:0] b'0000000 ~ b'1111111.

Step8. Loop Step7 GPIO1 single into find closest 3.125Mhz ~ 3.135MHz and record DV_TRIM_OSC value to file.

3.2 OTP Flow

Step1. [Save all result to file.](#)

Step2. Register write 28h.

Step3. Register write 10h.

Step4. Read register 0Ah 1Byte, check value(08h) on sleep in status.

Step5. Start OTP Flow: VPP Power on 7.5V

Step6. Register write 85h, 03h and wait >300us.

Step7. Register write 82h, 02h, 00h, 00h, 10h //OTP Program Group 2

Step8. Wait 10ms.

Step9. Register read 83h 2Byte, if read out value 00h,08h go to Step11 than go to Step9.

Step10. Register write 85h, 00h.

Step11. Register power off.

3.3 Re-Power on and OTP Reload

Step1. All power on.

Step2. Pin Reset

Step3. Register read 83h 5Byte if P2 G2_WC[1:0] of them is 0 abnormal.

Step4. Register Read 90h 3Byte if value is not default value, OTP reload finish.

4. Revision History

Version No.	Date	Page	Description
AN-K0018-250715	2025/7/15	5	Initial Version